D5. Retrieval Evaluation and Case Analysis

Measured retrieval quality with representative successes, failures and limitations.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Retrieval ablation result

Retrieval setup	Accuracy	Grounding	Recall@8	MRR	Reading
Dense-only	50.00%	87.50%	87.50%	0.3693	Ranks labelled chunks better but lower answer quality
Hybrid dense + BM25	62.50%	100.00%	87.50%	0.2781	Selected practical configuration

Case analysis method

Retrieval is assessed independently from the generated answer. The harness records each returned chunk ID and compares it with the stored gold chunk IDs for the unchanged structure-aware boundaries. This gives 87.50% Recall@8: evidence was missed for two of sixteen labelled cases. [E5][E10]

The dense-only result shows why no single ranking metric is enough: it produced stronger MRR/nDCG but weaker answer accuracy and grounding. Hybrid retrieval is selected for the current POC because it better supports the end-to-end assessment outcome, while future experiments should add independently annotated evidence spans. [E10]

Evaluation setup

The strongest saved retrieval evidence is the two-report production-parser run: 16 source-derived questions, 22 structure-aware chunks and top_k=8. Gold chunk identifiers are recorded in the run configuration. [E4][E6]

Metrics are reported independently of the answer model so retrieval limitations are not hidden by answer accuracy. [E5]

Measured retrieval metrics

Metric	Value	Interpretation
Precision@8	0.1443	Most top-8 context is not gold-labelled; evidence packing can improve.
Recall@8 / Hit Rate@8	0.8750 / 0.8750	Gold evidence retrieved for 14 of 16 labelled cases.
MRR	0.2781	Relevant evidence is frequently not ranked near first.
nDCG@8	0.4212	Ranking quality leaves material room for improvement.
Qwen3 comparison recall@5	1.0000	On the separate one-report/12-question, 10-chunk benchmark; not directly comparable to the two-report run.

Representative cases

Case type	Observed behaviour	Implication
Finding existence / severity / CWE	Many questions retrieve their labelled finding chunk.	Structured field paths aid local factual questions [E4][E6].
Negative or universal statement	Three of four No labels were predicted Yes by the pipeline.	Need negative/abstention examples and contradiction-aware retrieval [E5].
Missed labelled evidence	2 of 16 queries had recall@8=0.	Inspect chunk boundaries, labels and query formulation before increasing k [E5].
Small baseline	Perfect rank metrics were recorded on a 10-chunk one-report comparison.	Do not infer generalisation from the easy, overlapping-label setting [E2].

Interpretation

High grounding does not repair weak ranking: it means the final cited text is within returned evidence under the implemented check. It is not a claim-entailment guarantee. [E5][E8]

The next retrieval experiment should freeze reports, questions and labels, then isolate chunk length/overlap, dense-only versus hybrid, reranker, parent-child expansion and graph retrieval.

Evidence

Full per-query metrics, including the two recall failures, are preserved in retrieval_metrics.json. [E5]

Question-level retrieval exhibits

Question / report	Gold evidence retrieved?	Observed implication
RQ1 detailed findings / 2-REPORT	Yes; rank 4	The correct evidence is available, but not first-ranked.
RQ7 timeline / 4-REPORT	Yes; rank 1	A report-structure question can be efficiently answered.
RQ1 detailed findings / 4-REPORT	No; Recall@8 = 0	A correct answer may not prove retrieval success; inspect evidence separately.
RQ5 remediation / 4-REPORT	No; Recall@8 = 0	Second concrete recall miss requiring chunk/query review.

Evaluation judgement

The retrieval result is mixed but informative: Recall@8 is 87.50%, while MRR is 0.2781 and nDCG@8 is 0.4212. The evidence is usually somewhere in the eight returned chunks, but often not near the top. This is a real limitation for an audit assessor because extra context increases the chance of the generator selecting an irrelevant statement. [E5]

Hybrid retrieval is retained only as the better end-to-end control configuration: it improved accuracy to 62.50% and grounding to 100.00%, compared with dense-only 50.00% and 87.50%. These figures do not prove it is universally best; they justify the current POC choice and define the next ablation. [E10]

Reviewer inspection protocol

For each benchmark question, the reviewer should inspect three records together: the source-derived gold chunk ID, the returned top-eight chunk IDs and scores, and the final answer/cited evidence. This distinguishes a retrieval failure from a generation failure. The run has both cases: RQ1 and RQ5 for 4-REPORT miss the labelled evidence, while the final score can still look strong. [E4][E5]

The reported precision is deliberately low because one or a few gold chunks are compared with eight returned candidates. It should therefore not be read as eight irrelevant passages; it is a ranking diagnostic that shows how much non-gold context was passed to the model. The next practical improvement is reranking or reduced evidence packing, evaluated on independently labelled evidence spans. [E5][E8]

The conclusion is appropriately cautious: retrieval is functional and measurable, not yet accurate enough to be invisible in the final decision. The submission preserves the exact failure cases so an evaluator can challenge the claimed score with primary output rather than accepting a summary metric. [E5]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.
- [E6] sample_data/{2-report.json,4-report.json,real_reports_benchmark_questions.json}: source reports and eight source-derived questions per report.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.